

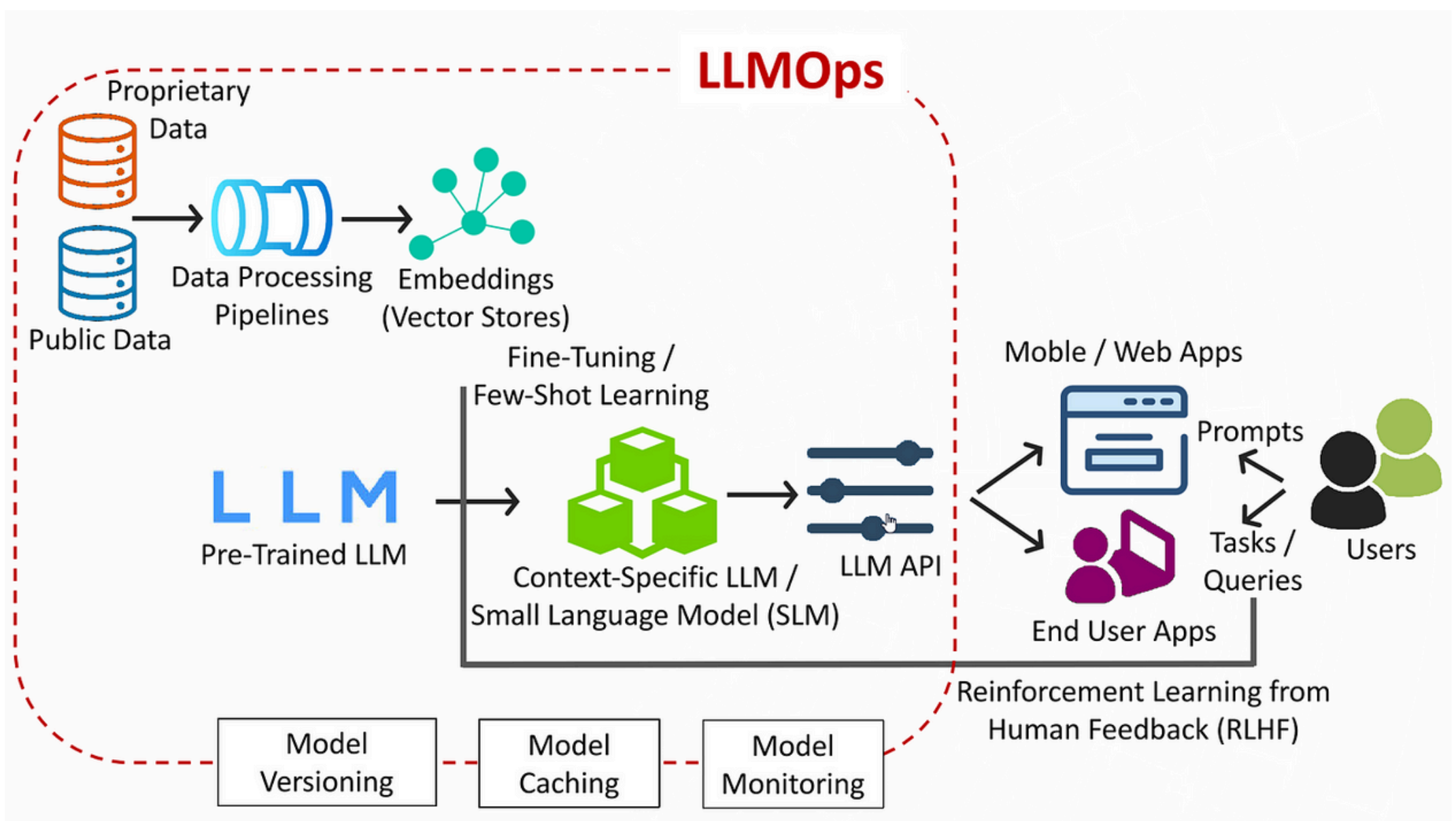
LLMOps

Productionalizing Real-World Applications with LLMs & Agents

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What is LLMOps?

- LLMOps stands for Large Language Model Operations a structured approach to building, deploying, and managing LLM-based systems in production environments.
- It ensures that GenAI applications go beyond prototypes and become scalable, observable, and reliable products.



At DataHack Summit 2025, we are
conducting a workshop on

 **Hack Session**

LLMOps – Productionalizing Real- World Applications with LLMs and Agents

Kartik Nighania

MLOps Engineer



Core areas of LLMOps

- Prompt and agent orchestration
- Retrieval-Augmented Generation (RAG) integration
- Model evaluation and feedback loops
- Automated deployment pipelines
- Monitoring, logging, and observability

Why LLMOps is Critical

Without LLMOps, LLM applications often suffer from:

- Hallucinated or factually incorrect responses
- Uncontrolled prompt and agent behavior
- Poor performance tracking and debugging
- Lack of scalability, automation, and reliability

LLMOps addresses these challenges through:

- Structured workflow design
- Continuous prompt and agent testing
- Evaluation of real-world performance
- Observability across latency, cost, and accuracy
- Deployment and monitoring pipelines

Core Components of a LLM System

Prompt and Agent Orchestration: Use frameworks like LangChain, LangGraph, or DSPy to define modular, multi-step workflows involving tools, memory, and conditional logic.

RAG Integration: Connect your LLM to vector databases (e.g., Qdrant, FAISS) to retrieve relevant and grounded information dynamically during inference.

Evaluation and Feedback: Implement structured evaluation with tools like LangSmith or Trulens to measure groundedness, factual accuracy, coherence, and user satisfaction.

Monitoring and Logging: Set up real-time logging, tracing, and alerting with observability tools such as Langfuse, Weights & Biases, or custom dashboards.

